

motors: <https://emaxmodel.com/products/emax-mt2213-935kv-multicopter-brushless-motor>



Framework:.....12N14P

KV:.....935KV

Length:.....39.7mm

Diameter:.....27.9mm

No.of cells:....3-4S

Max. thrust:....720g

Shaft:.....3mm

Propeller:.....8"-10"

Weight:53g

The voltage (V)	Paddle size	current (A)	thrust (G)	power (W)	efficiency (G/W)	speed (RPM)
11.1	EMAX8045	1	110	11.0	10.0	3650
		2	200	22.0	9.1	4740
		3	270	33.0	8.2	5540
		4	330	44.0	7.5	6200
		5	380	55.0	6.9	6700
		6	420	66.0	6.4	7150
		7.3	470	81.0	5.8	7400
	EMAX1045	1	130	11.0	11.8	2940
		2	220	22.0	10.0	3860
		3	290	33.0	8.8	4400
		4	360	44.0	8.2	4940
		5	430	55.0	7.8	5340
		6	490	66.0	7.4	5720
		7	540	77.0	7.0	5980
		8	580	88.0	6.6	6170
		9	620	99.0	6.3	6410
		9.5	640	105.5	6.1	6530
14.8	EMAX8045	1	130	14.8	8.8	3900
		2	230	29.6	7.8	5180
		3	310	44.4	7.0	6000
		4	390	59.2	6.6	6610
		5	470	74.0	6.4	7200
		6	530	88.8	6.0	7570
		7	580	103.6	5.6	7910
		8	630	118.4	5.3	8230
		9	670	113.2	5.0	8500
		10	700	148.0	4.7	8780
		10.7	720	158.4	4.5	9030

cog = 0.09 of tail starting from front ie where the horizontal line of the T is. Mass properties of frame assembly v2 Configuration: Default Coordinate system: -- default --
Mass = 1123.53 grams Volume = 578139.18 cubic millimeters Surface area = 353200.79 square millimeters Center of mass: (millimeters) X = -124.04 Y = -88.50 Z = 82.33
Principal axes of inertia and principal moments of inertia: (grams * square millimeters)
Taken at the center of mass. Ix = (1.00, 0.00, 0.00) Px = 7227994.11 Iy = (0.00, 0.02, -1.00) Py = 21297774.26 Iz = (0.00, 1.00, 0.02) Pz = 25912650.24 Moments of inertia: (grams * square millimeters) Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.) Lxx = 7228403.16 Lxy = -84692.42 Lxz = 18405.79 Lyx = -84692.42 Lyy = 25910627.99 Lyz = -87020.83 Lzx = 18405.79 Lzy = -87020.83 Lzz = 21299387.46 Moments of inertia: (grams * square millimeters) Taken at the output coordinate system. (Using positive tensor notation.) Ixx = 23643135.91 Ixy = 12248708.03 Ixz = -11455239.40 Iyx = 12248708.03 Iyy = 50812718.42 Iyz = -8273007.27 Izx = -11455239.40 Izy = -8273007.27 Izz = 47385519.92 propellers: 10 inch Pitch 4.5